

Angel Causal Boundary Modular SDK 1.2.0

Algorithm, State-Integrity, and Publication Audit

Annals Revision of 26 August 2026

Audit verdict

The reviewed Wilson arithmetic, three-class factorial bridge and cyclic identities are correct within the stated model. The implementation passes independent differential and boundary checks after twelve material audit repairs. The main residual limitations are the surviving square-root coordinate, fixed-word optimized modulus chart, and the absence of generalized noncentral HDIV and a compact native cyclic compiler.

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1 Executive conclusion

Outcome

No reviewed counterexample invalidated the q-Pochhammer concatenation, principal-jet factorial, complement-pairing, Wilson, divisor-kernel, primitive-projector, closure-valuation, quotient-transport, or scoped residual-horizon mathematics. The executable residue path agrees with independent arithmetic oracles in the audited domain and preserves the declared no-feedback boundary.

The original package nevertheless needed revision before it could support a mathematical-annals presentation. The largest issue was architectural: theorem/proof, finite test evidence, CI facts, hashes and release bookkeeping were interleaved at one narrative level. The expanded color volume exceeds 140 pages, gives the complete Angel Prime theorem chain, and places difficult proof expansions in a technical appendix and every finite implementation claim in evidence appendices.

2 Algorithm assessment

2.1 Factorial source and principal jet

The source polynomial

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j)$$

has the exact concatenation law

$$F_{a+b}(Z, u) = F_a(Z, u)F_b(Zu^a, u).$$

At $Z = -(1 + T)$ and $u = 1 + T$, the first nonzero coefficient is

$$(-1)^n n! T^n.$$

The implementation represents the binary source grammar and a certified factorial interval; it does not count the short source as a completed ordinary factorial download.

2.2 Wilson observer

For $m = n + 1$ and $h = \lfloor n/2 \rfloor$,

$$n! \equiv (-1)^h (h!)^2 (h + 1)^{n \bmod 2} \pmod{m}.$$

This identity is integral and valid for prime or composite m . The modular consumer obtains n from the certified native factorial rank and uses the candidate only as modulus and as the independent check $m = n + 1$.

The block evaluator is composite-safe because auxiliary-prime transforms reconstruct bounded integer convolution coefficients before reducing modulo the target, and remainder divisors are monic. No target-field inverse is assumed.

2.3 Cyclic chart

After characteristic-zero scalar extension, exact-period sectors diagonalize the cyclic self-action. This gives the divisor-stratified kernel, primitive-period projector, closure valuation

$$\text{ord}_\varepsilon(\mathcal{F}_{m,N}|_{V_d}) = \frac{Nm}{d},$$

and the time-valuation product $\tau_m(d)\nu_m(d) = m$. The mathematics is sound as a structural chart; the supplied cyclic evaluator remains a bounded reference implementation.

2.4 Three-class Angel arithmetic and functors

The revised public layer represents an exact value as

$$X = (x_{\text{Ord}}, x_{\text{Hist}}, x_{\text{Sing}}),$$

where ordered history words form an associative algebra and the singular sector is a two-sided ideal. Addition and subtraction are coordinatewise. HMUL is the unitized ordered-word product, so it is generally noncommutative outside the ordinary sector. Central-scalar HDIV returns an exact quotient and typed residual satisfying

$$X = \iota(d) Q_d(X) + R_d(X),$$

including $Q_0(X) = 0$ and $R_0(X) = X$. Unsupported noncentral division or checked coefficient overflow produces a continuation retaining its operands.

Native programs act functorially: the empty instruction word is identity and program concatenation is sequential composition with the complete event path and state seal retained. Class-quantum superposition is a finite integer-weight module on deterministic paths. Interference aggregates only structurally equal endpoints, exact opposite weights cancel, and entanglement is certified by a nonzero exact 2×2 coefficient minor. These are deterministic history semantics, not a physical quantum-mechanics claim.

2.5 Factorial maze bridge

The revised SDK now binds the generic three-class maze to the principal-jet factorial engine. Upload records argument, program, native-state and request words; execution independently verifies the native coordinate and appends one result/certificate word. Only the opaque executed carrier may enter exact materialization or joint Wilson observation. Two new negative-compilation programs prove that a download cannot re-enter execution and that upload cannot skip the execute stage.

2.6 Volume V research boundary

The integrated paper also presents the Causal Horizon Zipper and its residual-packet stability theorem. That chart has a separate strict research verifier and remains outside the public modular SDK. Its positive claim is amortized reuse of already-paid source work; it is not a $p^{o(1)}$ cold next-prime compiler.

2.7 Higher-dimensional structural probes

The post-algorithm research harness executes the optimized Wilson consumer on every candidate $2 \leq m \leq 4096$ and compares its frozen result, offline, with independent Euler- φ , divisor-support, cyclic-row and primality data. Separate modes directly audit 255 cyclic shadows, 547 period-rank cases and 64 higher factorial jets. The resulting 4,961 records are implementation evidence; the procyclic, Ramanujan, Laplacian and higher-jet statements have independent proofs in the mathematical volume.

3 Material findings and repairs

1. Baseline coordinate undercount

The reconstructed baseline owns both an evaluation-point array and a block-value array. The former formula charged one block-sized array and mixed streamed tail factors into stored space. The corrected definition is

$$C_0 = s_0 + 1 + 2\lfloor n/s_0 \rfloor.$$

2. Polynomial fast-path coverage

Transform/CRT multiplication and fast monic remainder were previously reached only indirectly. A dedicated test now compares them with independently written schoolbook references at composite and near-maximum 64-bit moduli.

3. Parallel equivalence

The default per-prime parallel transform path now has a direct coefficient-by-coefficient equality test against serial execution.

4. Derived-observation authority

The public `validate_download_packet` interface compares a transported packet with canonical closure. A tampered primitive column is rejected, and the test verifies that canonical state remains unchanged.

5. Publication role separation

The annals body now emphasizes definitions, examples, theorem statements and proof roadmaps. Full difficult proofs and every finite software claim appear only after the appendix boundary.

6. Missing unified high-dimensional arithmetic layer

The source volumes already contained unitized history arithmetic, singular quotients, causal histories and path packets, but the release did not expose one three-class axiom system, all four operations, or high-dimensional functors as a single executable contract. This revision adds ten explicit axioms, HADD/HSUB/HMUL/HDIV, native/history/future-quotient functors, class-quantum superposition/interference/cancellation, exact tensor entanglement certificates, and a public finite sparse implementation with dedicated positive and negative tests.

7. Missing factorial-to-maze correspondence

The earlier article placed the general three-class algebra before the factorial application without a theorem or API connecting them. The new factorial-maze chart, specialized high-to-high functor, bridge certificate, positive tests and two compile-time barriers close that gap while leaving the sealed principal-jet core unchanged.

8. Angel Prime transition scope was implicit

Wilson observation decides a supplied candidate and cannot explain native successor transport alone. The paper now integrates the strict Volume V source-first child and residual-horizon mathematics, labels it verifier-only, and states the remaining cold collective-source interface explicitly.

9. The larger object had no falsifiable finite probes

The factorial and cyclic charts suggested a much larger atlas, but the release did not cross-check their finite shadows through the optimized Wilson path. The new additive research harness records complete Wilson, primitive-rank, Ramanujan-row, divisor-support and higher-jet campaigns without feeding any offline result back into native execution.

10. Three sectors were mistaken for three spatial axes

The shorter exposition did not distinguish semantic sector count from module rank and did not state the duties of Room, Road, Frame, History, Holonomy, Residual and HigherCell. The revision constructs a free exact maze category, defines every coordinate's role and proves a scoped minimal-separation result.

11. Binary programme diagram skipped required states

Two append edges were drawn as direct jumps $2 \mapsto 5$ and $10 \mapsto 21$. The corrected figure restores the required doubles $2 \mapsto 4$ and $10 \mapsto 20$, so the picture now agrees with the verified bit recurrence.

12. Origin and predecessor history were compressed inaccurately

The naming paragraph had replaced the author's reported dream, workday lunch-nap recognition and forgotten-formula boundary by a different anecdote. The revision records that account only as authorial provenance, restores the SN-OFCS precursor map, proves typed zero-division conservativity, and states that the present q-Pochhammer atlas is a proved reconstruction rather than a verbatim recovery of the lost formula.

4 Claim boundary

Scope boundary

The audit does not certify a polylogarithmic Wilson consumer, elimination of the $\sqrt{n}$ coordinate, an arbitrary-precision optimized modulus path, a competitive general primality test, or a descriptor-native subpolynomial cold next-prime compiler. The finite Pareto rows are implementation evidence, not an asymptotic theorem. Nor does the finite SDK claim arbitrary-precision coefficients, generalized inversion of every noncentral high-dimensional

denominator, or physical quantum dynamics.

A Executable evidence

Campaign	Acceptance condition	Result
Debug/release	complete transcripts agree	PASS
ASAN/UBSAN	all positive suites complete	PASS
Negative compilation	14 prohibited programs fail to compile	PASS
Frozen/API hashes	protected sources and legacy headers agree	PASS
Three-class algebra	HADD/HSUB/HMUL laws and singular-ideal retention agree	PASS
HDIV reconstruction	quotient plus typed residual reconstructs the numerator	PASS
Causal functors	identity and composition preserve full state and history	PASS
Class-quantum paths	structural interference and exact cancellation agree	PASS
Entanglement	separability and a nonzero exact minor certificate agree	PASS
Exact factorial	two independent big-integer derivations agree	PASS
Factorial maze	tri-class binding, functor bridge and both executed consumers agree	PASS
Polynomial differential	transform/CRT and remainder equal schoolbook oracles	PASS
Parallel/serial	per-prime transform schedules agree	PASS
Derived-map tamper	modified packet rejected; canonical state unchanged	PASS
Continuous Pareto	work, peak and coordinate inequalities hold	PASS
Volume V research verifier	zipper/cold child hashes and residual stability agree	PASS*
Wilson structural atlas	4,095 candidate rows agree with primality, φ and divisor support	PASS
Cyclic shadow probe	pure Laplacian exactly on 54 prime rooms among 255; all 54 prime row folds give -1	PASS
Period-rank filtration	547 modular matrix ranks equal the divisor-sector formula	PASS
Higher factorial jets	64 principal and first higher identities agree	PASS

*Separate strict research verifier; not an exported SDK API.

B Corrected deterministic sequence

All ratios are below one. Their scope is the fixed sequence and published counter model. The exact machine-readable rows are archived in `results/joint_wilson_operation_counts_annals_revision.csv`.

m	W_1/W_0	P_1/P_0	C_0	C_1	C_1/C_0
257	.4746	.3256	49	9	.1837
1021	.4701	.3333	95	18	.1895
4093	.4490	.3260	191	41	.2147
16381	.4487	.3285	383	91	.2376
65521	.4441	.3301	767	181	.2360
100003	.4427	.3281	948	224	.2363
250003	.4412	.3301	1500	354	.2360
500009	.4436	.1917	2121	501	.2362
1000003	.6417	.5280	3000	708	.2360

Table 2: Regenerated finite ratios after coordinate-accounting correction.

C Reproduction and limitations

```
./build_and_test.sh
cmake -S . -B build -DANGEL_BUILD_TESTS=ON
cmake --build build --parallel
ctest --test-dir build --output-on-failure
```

The annals revision build records identical debug/optimized transcripts, passing address and undefined-behavior sanitizers, passing negative compilation and compatibility checks, the polynomial differential, parallel equivalence and map-validation checks, and the new three-class arithmetic, functor, class-quantum, entanglement and factorial-maze bridge checks, plus the Wilson/cyclic/rank/higher-jet structural campaigns. Leak detection is marked skipped in the ptrace-restricted environment.

The remaining engineering work is a general L -bit optimized modulus chart, generalized noncentral HDIV, elimination or lower-bound analysis of the square-root coordinate, and a compact proof-carrying cyclic compiler. All-chart reconstruction and native Angel Prime birth remain mathematical open interfaces, not implied by the finite structural campaign.